A Novel MIMO-SAR Echo Separation Solution for Reducing the System Complexity: Spectrum Preprocessing and Segment Synthesis

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***Abstract*— The problem of echo separation using digital beamforming (DBF) on receive for multiple-input multiple-output (MIMO) synthetic aperture radar (SAR) is of notable importance to allow for practical systems. Regrettably, current DBF-MIMO-SAR schemes, such as the short-term shift-orthogonal (STSO) scheme, are computationally cumbersome, increasing the required hardware complexity. To alleviate this problem, we here propose an improved echo separation solution for realizing a low-cost MIMO-SAR system. We detail a generic waveform design scheme and optimized monostatic radar waveforms (e.g., nonlinear frequency modulation (NLFM) signal), showing how these can be directly adopted in the proposed scheme to improve imaging performance. The proposed scheme enables the number of interference segments generated by unmatched waveforms to be halved by the use of the fast time spectrum preprocessing and segment synthesis, dramatically simplifying the array configuration, and reduces the system complexity. By exploiting interpulse phase coding techniques, the proposed method can provide a reconfigurable waveform transmitting scheme, allowing the system resources in range frequency, elevation space, and Doppler domains to be jointly exploited for the separation of aliased signal returns. The proposed scheme is evaluated using extensive numerical and measured datasets, demonstrating the feasibility and potential of the proposed method for resource-limited spaceborne/airborne MIMO-SAR systems.**

***Index Terms*— Digital beamforming (DBF), interpulse phase coding, resource-limited multiple-input multiple-output (MIMO) synthetic aperture radar (SAR), segment synthesis, spectrum preprocessing.**

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I. Introduction

MULTIPLE-INPUT multiple-output (MIMO) systems were introduced in wireless communication in the early 1990s [1], and the concept has recently been applied to synthetic aperture radar (SAR) in order to improve the system imaging performance through waveform and spatial diversities [2]. With the aid of multiple transmit and receive antennas, the forthcoming generation of SAR instruments, often termed MIMO-SAR, can circumvent the fundamental antenna configuration constraint limiting conventional SAR systems and, thereby, achieve considerable improvement in areas such as high-resolution wide-swath (HRWS) imaging [3], downward-looking 3-D surveillance [4], and ground moving target indication (GMTI) [5], [6]. As a result of these improvements, MIMO-SAR has attracted notable interest for both military and civilian applications in recent years.

A. Background

At each receiver in an MIMO-SAR system, the recorded signals are composed of matched desired signals and unmatched interferences (unfocused responses to the transmitted *primary* waveform). Typically, the interference energy will be widely spread in the time–frequency plane, and unlike in communication systems, this energy cannot be neglected in an SAR system due to the potential presence of distributed targets [7], [8]. Therefore, it is of notable importance to ensure a channel response separation able to fully exploit the benefits provided by the MIMO configuration. Generally, the waveform orthogonality, which can be acquired by the use of, e.g., time-division multiplexing (TDM) [9] or frequency-division multiplexing (FDM) [10], helps to facilitate the required separation processing. Nevertheless, the scattering response of a distributed reflection is usually time-varying and/or frequency-selective, resulting in potential performance degradation for both TDM and FDM systems (assuming a coherent operation) [11]. Furthermore, the used waveforms must be physically attainable and amenable to a real transmitter [12]. To date, there are no perfectly time and frequency synchronized orthogonal waveforms that are realistic to implement using real-life hardware [13]. The echo separation conundrum exemplifies the practical difficulty encountered

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with constructing a realistic MIMO-SAR system and has also highlighted the need for further research on new methods that are able to separate waveforms with the same time and frequency support.

B. Previous Work

Recently, contributions on the use of the Alamouti space–time coding (STC) scheme have been reported to allow for the elimination of the cross correlation energy (CCE) [14]. The STC principle is also evaluated by airborne experiments [15]. However, the use of an STC spatial diversity scheme would lower the amount of available spatial samples in azimuth, which are crucial for HRWS imaging or GMTI [16]. For this reason, sophisticated orthogonal-waveform beamforming schemes, e.g., the short-term shift-orthogonal (STSO) scheme [2] and the segment phase coding (SPC) scheme [17], [18], have attracted considerable attention. With the employment of either of these particular modulation waveforms, an MIMO-SAR system, in combination with digital beamforming (DBF) on receive in elevation [19], can avoid loss of imaging performance [2]. Furthermore, in contrast to the STSO waveform proposed by Krieger [2], SPC reduces the need for the waveform orthogonality in the STSO scheme and, thereby, increases the number of suitable radiation waveforms. The SPC scheme has been demonstrated by an airborne DBF-SAR system [20].

Although both the STSO and the SPC schemes can yield satisfactory echo separation performance, the use of DBF with the well-established Tx/Rx module techniques will substantially increase the complexity of the MIMO-SAR system [21]. To separate the aliased signal returns from the M transmitted waveforms, at least $2M-1$ subapertures need to be arranged in elevation, necessitating a large antenna array with many digital elements. Since the design, construction, and deployment of remote sensing systems with multiple digital elements are rather expensive [10], [22], this limits the usability of these MIMO-SAR systems, reducing the possibility of constructing low-mass and low-cost airborne or unmanned aerial vehicle (UAV) SAR sensors that have been widely investigated in recent years [23].

To simplify the required array configuration and relax the hardware requirements of MIMO-SAR systems, the sequence CLEAN deconvolution technique, initially developed for radio astronomy [24], has been examined for the separation of signal returns [25]. The sequence CLEAN method enables the echo separation without any system cost although the increased computation complexity, requiring multiple match filtering iterations for separating the M waveforms, may make the system slower [25]. In addition, the CCE in the distributed scene cannot be thoroughly suppressed by the CLEAN technique even though the image quality is moderately improved [26]. Overall, the development of an effective waveform separation scheme for a practical remote sensing system with limited available channels [i.e., the limited degrees of freedom (DOFs)] is recognized as the most challenging issue to be resolved in order to allow for a practical MIMO-SAR system.

C. Main Contribution and Organization

Considering the abovementioned technical challenges, we here introduce a pragmatic echo separation solution based on the time–frequency properties of STSO waveforms. The main contributions of our work are given as follows.

1) The proposed technique strives to fully exploit the time–frequency properties of the STSO scheme. With the employment of a bandpass filter bank (each filter being centered over the bandwidth of a different signal segment in the range frequency domain), a separation of the aliased signals radiated by M STSO waveforms is achievable by the proposed technique even when only M elevation subapertures are deployed in MIMO-SAR system, in clear contrast to the $2M-1$ subapertures required by the current elevation beamforming schemes. This improvement considerably reduces the system complexity and notably affects the possibility to produce low-cost and low-mass SAR sensors, being necessary to allow for the employment of such sensors in small SAR satellites.
2) For elevation-space-limited MIMO-SAR systems, the echo separation capability is often inadequate. To alleviate this, we here introduce the azimuth phase code (APC) technique to implement the echo separation in the Doppler domain [27]. In particular, the collaborative processing in range frequency, elevation space, and Doppler domains does not depend on the availability of a large antenna array or the use of a high pulse repetition frequency (PRF). Sufficient operational flexibility and reconfigurability can instead be obtained by specifically arranging the encoding scheme, which is especially important for the multimodal operation of MIMO-SAR.
3) A preliminary MIMO system (with two transmitters and two receivers) has been developed in our work. Several practical aspects of the resulting system have also been addressed, including imperfection and implementation issues, such as the presence of phase, gain, and offset mismatch. The developed MIMO system provides an opportunity to evaluate the current waveform transmission schemes and their echo separation performance in a more practical setting.

The remainder of this article is organized as follows. In Section II, a generic multiple-subpulse time-shifting (MSTS) waveform design is detailed, and we introduce the waveform transmission scheme of the proposed method. Section III gives an introduction to the echo separation solution combining interpulse phase decoding and beamforming in the elevation spatial domain. In Section IV, we proceed to examine some important practical issues, such as range resolution, system cost, and computational complexity, illustrating the relative merits and tradeoffs concerning the proposed method. In Section V, comprehensive simulations are carried out to prove the effectiveness of the proposed method, and the developed experimental MIMO-SAR system is also introduced. Finally, Section VI provides concluding remarks.

Notations: Boldface uppercase and lowercase letters denote matrices and vectors, respectively. $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$

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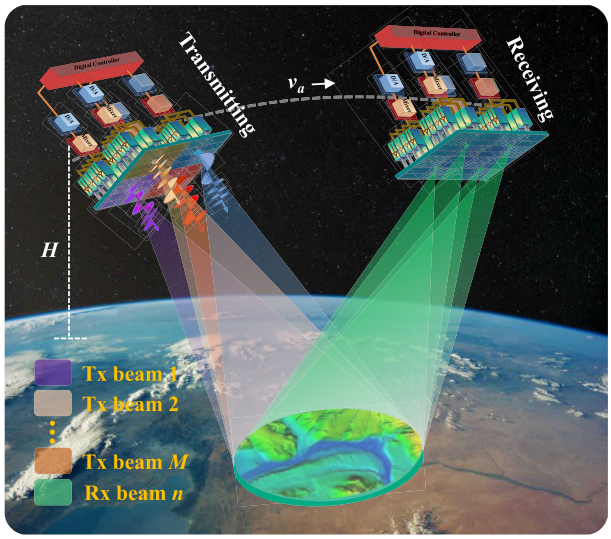


Fig. 1. Geometric reception model for an MIMO-SAR system. Here, multiple transmitters are closely spaced, and the target's scattering response is the same for each antenna pair [11].

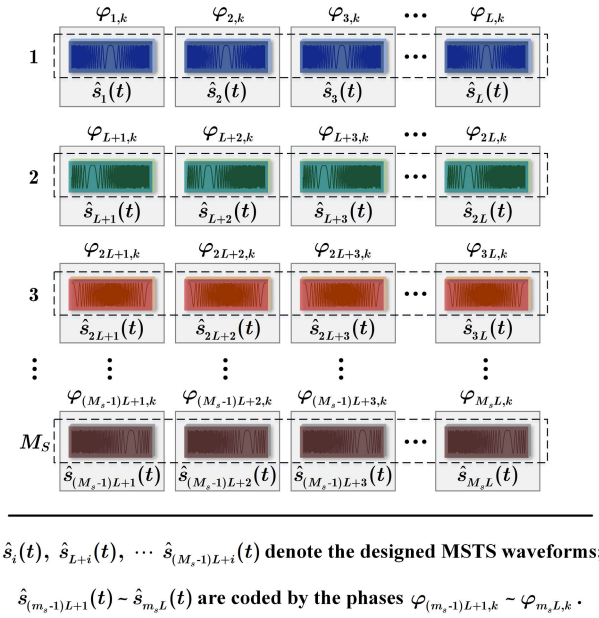


Fig. 2. Waveform design for the proposed method. M_s MSTS waveforms (marked by the different colors) are designed based on the STSO beamforming principle, while L waveforms (characterized by the same color) are coded using the APC technique, with the total number of the radiated waveforms satisfying $M \leq M_s L$.

are the complex conjugate, transpose, Hermitian transpose, and inversion, respectively. $\otimes_t$, $\mathcal{F}_t(\cdot)$, and $\mathcal{F}_t^{-1}(\cdot)$ represent the convolution, the Fourier transform, and the inverse Fourier transform in the fast time domain, respectively. $\mathcal{F}_\eta(\cdot)$ and $\mathcal{F}_\eta^{-1}(\cdot)$ are the Fourier transform and inverse Fourier transform across the azimuth direction, respectively. $|\cdot|$ and $\|\cdot\|$ refer to the absolute value and the modular operation, respectively. $\mathbf{0}$ indicates the zero matrix. $\mathrm{rank}(\mathbf{A})$ represents the rank of matrix $\mathbf{A}$. $\Omega\{\mathbf{A}(i,:)\}$ returns the value of j if $\mathbf{A}(i,j)=1$. $\lfloor\cdot\rfloor$ and $\lceil\cdot\rceil$ indicate the rounding down and rounding up operations, respectively. $\min(a,b)$ returns the smaller value. $\arcsin\{\cdot\}$ indicates the arcsine function. $\odot$ represents the Hadamard (elementwise) product. Finally, $\mathbb{R}^{M\times N}$ and $\mathbb{C}^{M\times N}$ refer to the matrix space representing $M\times N$ over a real number field and complex field, respectively.

II. Transmitting Signal Model for MIMO-SAR

The design of suitable transmission waveforms plays an important role in the development of an MIMO-SAR system. In this section, we briefly introduce the coding scheme employed by the proposed method. Fig. 1 shows a generalized imaging geometry for an MIMO-SAR system with M isotropic transmitters. The transmitters and receivers (denoted by Txm and Rxn, respectively) in this system are all near colocated, deployed on a radar platform with a height H. It is worth noting that, in the go-stop model [17], Txi and Rxi occupy the same spatial position. The sensors are assumed to move in the along-track direction (X-axis direction) with a constant forward velocity, v_a. As shown in Fig. 1, the backscattered radar signals, simultaneously radiated by M transmitters, are collected by the subapertures, which are connected to the N individual receiving channels. Generally, the receiving antennas in elevation are employed to perform DBF processing (for the separation of channel responses) instead of acquiring more effective phase centers (EPCs).

To achieve desired echo separation and to acquire better imaging performance with the limited system resources, the APC technique, in combination with the improved elevation beamforming scheme, has been exploited due to its negligible implementation complexity [28], [29], [30]. The schematic of the waveform encoding model is shown in Fig. 2. First, the generic form of MSTS waveforms is detailed. Inspired by Krieger [2], to achieve an optimal DBF performance (see Section III), the time interval that ensures the orthogonality should be maximized (for M_s STSO waveforms, the optimal time interval is equal to T_p/M_s, where T_p refers to the pulsewidth). Typically, similar to the STSO scheme, the MSTS waveforms are composed of linear- or cyclic-shifted subpulses, allowing the extended partially orthogonal waveforms to be established by a set of radar signal bases, which are expressed as

$$\begin{aligned} \boldsymbol{s}_0(t) &= \left[w_1(t)\, s(t), w_2(t)\, s(t), \ldots, w_{M_s}(t)\, s(t)\right]^T \\ w_{m_s}(t) &= \mathrm{rect}\left(\frac{t - \frac{2m_s - M_s - 1}{2M_s} T_p}{T_p/M_s}\right), \quad 1 \le m_s \le M_s \end{aligned} \tag{1}$$

where t represents the fast time and $s(t)$ represents the monostatic radar signal, e.g., linear frequency modulation (LFM) [31], nonlinear frequency modulation (NLFM) signals [32], [33], or the optimized phase coding waveforms (such as polyphase-coded frequency modulation (PCFM) [34], [35]). Without loss of generality, we here employ the NLFM waveform as a good alternative to the LFM waveform, given its low-range sidelobes, and the fact that it is physically realizable by power-efficient transmitters. The rectangular window function

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$\mathrm{rect}(t)$ in (1) is defined as

$$\mathrm{rect}\,(t)=\begin{cases}1, & -0.5\le t\le 0.5\\ 0, & \text{else.}\end{cases} \tag{2}$$

With the signal base $\boldsymbol{s}_0(t)$, the designed MSTS waveform is here formed as

$$\hat{s}_{m_s}(t)=\sum_{i=1}^{M_s}\left(\boldsymbol{e}_{m_s,i}^{\gamma_i}\right)^T\boldsymbol{s}_0\left(t+\frac{\gamma_i-i}{M_s}T_p\right) \tag{3}$$

where $\boldsymbol{e}_{m_s,i}^{\gamma_i}\in\mathbb{R}^{M_s\times 1}$ denotes the unit vector formed as

$$\boldsymbol{e}_{m_s,i}^{\gamma_i}=\left[\underbrace{0,\ldots,0}_{\gamma_i-1},1,0,\ldots,0\right]^T,\quad i,\gamma_i=1,2,\ldots,M_s. \tag{4}$$

To allow the range spectrum of $\hat{s}_{m_s}(t)$ to occupy the same bandwidth as $s(t)$, the unit vectors in (4) should be orthogonal with each other, i.e.,

$$\left(\boldsymbol{e}_{m_s,i}^{\gamma_i}\right)^T\boldsymbol{e}_{m_s,j}^{\gamma_j}=0,\quad 1\le i,j\le M_s,\quad i\ne j. \tag{5}$$

For notation simplicity, we define a nonsingular binary matrix (with entries of either 0 or 1) to represent the unit vectors such that

$$\mathbf{E}_{m_s}=\left[\boldsymbol{e}_{m_s,1}^{\gamma_1},\boldsymbol{e}_{m_s,2}^{\gamma_2},\ldots,\boldsymbol{e}_{m_s,M_s}^{\gamma_{M_s}}\right]\in\mathbb{R}^{M_s\times M_s}. \tag{6}$$

In addition, to make the echoes from different Txs separable in the elevation spatial domain, the modulation matrices $\mathbf{E}_{m_s}$ for different transmitting waveforms also need to satisfy [2]

$$\mathbf{E}_i\odot\mathbf{E}_j=\mathbf{0},\quad 1\le i,j\le M_s,\quad i\ne j. \tag{7}$$

By combining (3) and (7), one may then design satisfactory transmitting waveforms for an MIMO-SAR system. Based on the modulation scheme in Fig. 2, the M_s transmitted baseband signals can be expressed as

$$\hat{s}_{(m_s-1)L+l}(t)=\sum_{i=1}^{M_s}\left(\boldsymbol{e}_{m_s,i}^{\gamma_i}\right)^T\boldsymbol{s}_l\left(t+\frac{\gamma_i-i}{M_s}T_p\right) \tag{8}$$

for $l=1,2,\ldots,L$, $m_s=1,2,\ldots,M_s$, with

$$\boldsymbol{s}_l(t)=\left[w_1(t)\,\bar{s}_l(t),w_2(t)\,\bar{s}_l(t),\ldots,w_{M_s}(t)\,\bar{s}_l(t)\right]^T \tag{9}$$

where $\bar{s}_l(t)$ represents an arbitrary NLFM radar signal with a continuous time–frequency curve. In a particular case, (8) may also be expressed as

$$\begin{aligned}&\hat{s}_{(m_s-1)L+l}(t)\\&=\begin{cases}\bar{s}_l(t), & m_s=1\\ w_r^{m_s,1}(t)\,\bar{s}_l\left(t-\dfrac{(M_s-m_s+1)T_p}{M_s}\right)\\ \quad+w_r^{m_s,2}(t)\,\bar{s}_l\left(t+\dfrac{(m_s-1)T_p}{M_s}\right), & \text{else}\end{cases}\end{aligned} \tag{10}$$

with $w_r^{m_s,1}$ and $w_r^{m_s,2}$ given by

$$\begin{aligned}w_r^{m_s,1}(t)&=\mathrm{rect}\left(\frac{t-\frac{(M_s-m_s+1)T_p}{2M_s}}{(m_s-1)T_p/M_s}\right)\\ w_r^{m_s,2}(t)&=\mathrm{rect}\left(\frac{t+\frac{(m_s-1)T_p}{2M_s}}{(M_s-m_s+1)T_p/M_s}\right).\end{aligned} \tag{11}$$

In the case of M_s=4, the modulation matrices corresponding to the waveforms in (10) are

$$\begin{aligned}\mathbf{E}_l&=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\qquad \mathbf{E}_{L+l}=\begin{bmatrix}0&0&0&1\\1&0&0&0\\0&1&0&0\\0&0&1&0\end{bmatrix}\\ \mathbf{E}_{2L+l}&=\begin{bmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{bmatrix},\qquad \mathbf{E}_{3L+l}=\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{bmatrix}.\end{aligned} \tag{12}$$

Including the APC modulation, the baseband signal for the transmission antenna m, at the kth pulse repetition interval (PRI), may be expressed as

$$\tilde{s}_{m,k}(t)=\varphi_{m,k}\hat{s}_m(t) \tag{13}$$

where m=1, 2, …, M, m and $k\in\mathbb{Z}$, with $\varphi_{m,k}$ denoting the interpulse phase modulation factor for the mth radiated waveform and the kth PRI. Assuming that the system PRF, denoted by F_D, $\varphi_{m,k}$ is defined as

$$\begin{aligned}\varphi_{m,k}&=\exp\left\{j2\pi\frac{F_D}{L}(m-1)\,k\cdot\mathrm{PRI}\right\}\\&=\exp\left\{j2\pi\frac{(m-1)\,k}{L}\right\}.\end{aligned} \tag{14}$$

Using the APC modulation, the Doppler bandwidth B_D of the returned signal must be smaller than F_D/L, i.e., $L\le\lfloor F_D/B_D\rfloor$. Otherwise, the returns experiencing aliasing in the Doppler spectrum cannot be completely separated using the bandpass filter bank. The echo signal measured at Rxn resulting from the modulated radar signal transmitted by the mth Tx may then be expressed as

$$r_{n,m}(t,\eta)=h_{n,m}(t,\eta)\otimes_t\tilde{s}_{m,k}(t) \tag{15}$$

where η represents the slow time scale referenced to the time of closest approach, i.e., $-\lfloor N_a/2\rfloor\le\eta\cdot F_D\le\lceil N_a/2\rceil-1$, with N_a denoting the number of azimuth sampling pulses. For an arbitrary point-like target P with complex reflectivity σ_0, the channel response $h_{n,m}(t,\eta)$ can be well approximated as

$$h_{n,m}(t,\eta)=\sigma_0\delta(t-t_{n,m})\exp(-j2\pi f_c t_{n,m}) \tag{16}$$

with f_c denoting the carrier frequency. Let the positions of Txm, Rxn, and the point target P be represented by $\boldsymbol{r}_{\mathrm{Tx}}^m$, $\boldsymbol{r}_{\mathrm{Rx}}^n$, and $\boldsymbol{r}_0$. Then, the round-trip time-delay $t_{n,m}$ (with reference to the geometric model in Fig. 1) is given by [31]

$$\begin{aligned}t_{n,m}&=\left(\|\boldsymbol{r}_{\mathrm{Tx}}^m-\boldsymbol{r}_0\|+\|\boldsymbol{r}_{\mathrm{Rx}}^n-\boldsymbol{r}_0\|\right)/c\\&\approx\left(2\sqrt{R_0^2+(v_a\eta)^2}+\Delta x_m\sin\theta_e+\Delta x_n\sin\theta_e\right)\Big/c\\&\approx\left(2R_0+\frac{v_a^2\eta^2}{R_0}+(\Delta x_m+\Delta x_n)\sin\theta_e\right)\Big/c\end{aligned} \tag{17}$$

with

$$\begin{aligned}\Delta x_m&=\{m-(M+1)/2\}\,L_e/M\\ \Delta x_n&=\{n-(M+1)/2\}\,L_e/M\end{aligned} \tag{18}$$

where c, R_0, θ_e, and L_e represent the speed of light, the closest slant range, the incident angle in the elevation, and the

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antenna height, respectively. Here, Δx_m and Δx_n denote the distance of the mth transmit subaperture and the nth receive subaperture to the array center, respectively.

Considering that each subaperture collects all the signal returns resulting from the different Txs, the measured aliased echoes of the nth Rx, at the kth PRI, can be described as

$$r_n(t,\eta)=\sum_{m=1}^{M} h_{n,m}(t,\eta)\otimes_t \tilde{s}_{m,k}(t)=\boldsymbol{h}_n^T(t,\eta)\otimes_t \tilde{\boldsymbol{s}}_k(t) \tag{19}$$

with

$$\begin{aligned}\boldsymbol{h}_n(t,\eta)&=\left[h_{n,1}(t,\eta),h_{n,2}(t,\eta),\ldots,h_{n,M}(t,\eta)\right]^T\\ \tilde{\boldsymbol{s}}_k(t)&=\left[\varphi_{1,k}\hat{s}_1(t),\varphi_{2,k}\hat{s}_2(t),\ldots,\varphi_{M,k}\hat{s}_M(t)\right]^T.\end{aligned} \tag{20}$$

The separation of the aliased signal returns in a single domain produces a considerable technical challenge, particularly for the multimodal operation of MIMO-SAR. However, the designed waveform transmission scheme makes the echo separation (implemented in the Doppler and elevation spatial domains) possible and is expected to allow for better imaging performance in most practical cases.

III. Echo Separation Scheme of Proposed Method

To fully exploit the potential benefits provided by the MIMO structure, this section focuses on the separation of aliased echo signals. Due to the designed waveform transmission scheme introduced in Section II, the echo separation can be divided into two parts: phase decoding in the Doppler domain and DBF processing in the elevation direction.

A. Azimuth Phase Decoding

First, the interpulse phase decoding operation is performed in this section. Given that the decoding is implemented in the Doppler domain, we form the fast Fourier transform (FFT) of (19) across the azimuth dimension, i.e.,

$$\begin{aligned}R_n(t,f_\eta)&=\sum_{m=1}^{M}\\ &\quad\times\mathcal{F}_\eta\left\{\exp\left(j2\pi\frac{F_D}{L}(m-1)\eta\right)h_{n,m}(t,\eta)\otimes_t\hat{s}_m(t)\right\}\\ &=\sum_{m=1}^{M}\int_{-\infty}^{\infty}h_{n,m}(t,\eta)\otimes_t\hat{s}_m(t)\\ &\quad\times\exp\left(j2\pi\left(\frac{F_D}{L}(m-1)-f_\eta\right)\eta\right)d\eta\\ &=\sum_{m=1}^{M}H_{n,m}\left(t,f_\eta-(m-1)\frac{F_D}{L}\right)\otimes_t\hat{s}_m(t)\end{aligned} \tag{21}$$

with $H_{n,m}(t,f_\eta)=\mathcal{F}_\eta\{h_{n,m}(t,\eta)\}$, and where f_η denotes the discrete Doppler frequency. It can be observed from (19) that the channel responses for all the Txs have different Doppler center deviations, i.e., $(m-1)F_D/L$. In addition, as mentioned in (14), the Doppler width B_D of $\hat{s}_m(t)$ is smaller than F_D/L, ensuring that there is no overlap between the adjacent azimuth frequency bands. Consequently, the aliased returns can be roughly separated by a set of bandpass filters in the Doppler domain. To filter out the echoes transmitted by Txm,

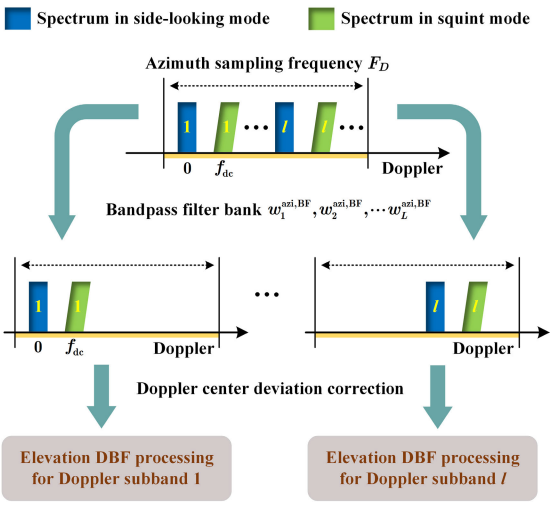


Fig. 3. Azimuth phase decoding for different work modes.

the bandpass filter can be formed using a rectangular window function with

$$w_m^{\mathrm{azi,BF}}(t,f_\eta)=\mathrm{rect}\left(\frac{f_\eta-f_{\mathrm{dc}}-(m-1)F_D/L}{F_D/L}\right) \tag{22}$$

where f_{dc} represents the Doppler center deviation of $\hat{s}_1(t)$. Generally, f_{dc} is considered to be zero for the side-looking SAR mode, while $f_{\mathrm{dc}}\neq 0$ for most squint cases, as illustrated in Fig. 3. Moreover, the Doppler center deviation caused by the phase coding needs to be corrected for subsequent imaging processing. The resulting separated echoes of $\hat{s}_m(t)$ can then be expressed as

$$\begin{aligned}\tilde{r}_{n,m}(t,\eta)&=\mathcal{F}_\eta^{-1}\left\{\left(w_m^{\mathrm{azi,BF}}(t,f_\eta)R_n(t,f_\eta)\right\}\right.\\ &\quad\times\exp\left(-j2\pi\frac{m-1}{L}k\right).\end{aligned} \tag{23}$$

As can be seen in (21), in the case of $L\geq M$, i.e., $M_s=1$, the aliased returns can be directly separated by (23). Otherwise, an additional beamforming operation should be implemented to acquire the single channel responses, as shown in Fig. 3.

B. DBF Processing in the Elevation Spatial Domain

Typically, the sampling frequency in azimuth is usually kept low to avoid range ambiguity and reduce the burden of the radar data downlink. Consequently, the resources in the Doppler domain are usually not enough to accomplish the separation of the M transmitted waveforms. In this situation, $L<M$ and $M_s\geq 2$, indicating that the echoes of multiple radar waveforms are superimposed in the same Doppler subband l, as illustrated in Fig. 3. For the lth subband, the recorded aliased signals are transmitted by the m_ith Tx, where

$$m_i\in\boldsymbol{\Theta}_l,\ \boldsymbol{\Theta}_l=\left[m_i|m_i=l+(m_s-1)L\right] \tag{24}$$

for $l=1,2,\ldots L$ and $m_s=1,2,\ldots,M_s$. Afterward, the collected MIMO-SAR data resulting from the aliased returns, in Doppler subband l, can be expressed as

$$\begin{aligned}\tilde{r}_n^{\boldsymbol{\Theta}_l}(t,\eta)&=\sum_{m_i\in\boldsymbol{\Theta}_l}\tilde{r}_{n,m_i}(t,\eta)=\sum_{m_i\in\boldsymbol{\Theta}_l}h_{n,m_i}(t,\eta)\otimes_t\hat{s}_{m_i}(t)\\ &=\boldsymbol{h}_{n,m_i}^T(t,\eta)\otimes_t\hat{\boldsymbol{s}}_{m_i}(t)\end{aligned} \tag{25}$$

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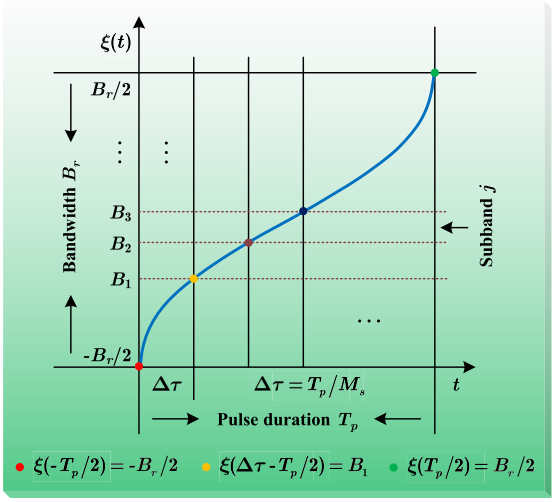


Fig. 4. Time–frequency function of the NLFM signal $\bar{s}_l(t)$. As can be seen, the bandpass filter widths for each subband are usually different.

where

$$\begin{aligned} \boldsymbol{h}_{n,m_i}(t,\eta) &= [h_{n,m_i}(t,\eta)]^T, \quad m_i \in \boldsymbol{\Theta}_l \\ \hat{\boldsymbol{s}}_{m_i}(t) &= [\hat{s}_{m_i}(t)]^T, \quad m_i \in \boldsymbol{\Theta}_l. \end{aligned} \tag{26}$$

To separate $\tilde{r}_{n,m_i}(t,\eta)$ from $\tilde{r}_n^{\boldsymbol{\Theta}_l}(t,\eta)$, the signal in (25) is filtered through a match filter, $\hat{s}_{m_0}(t), m_0 \in \boldsymbol{\Theta}_l$, yielding

$$\begin{aligned} \bar{r}_n^{\boldsymbol{\Theta}_l}(t,\eta) &= \tilde{r}_n^{\boldsymbol{\Theta}_l}(t,\eta) \otimes_t \hat{s}_{m_0}^*(-t) \\ &= \sum_{m_i \in \boldsymbol{\Theta}_l} \sum_{j=1}^{M_s} h_{n,m_i}(t,\eta) \otimes_t p_{m_0,j}\left(t + \frac{\beta_{m_i,j}}{M_s} T_p\right) \\ &= \bar{r}_{n,\mathrm{des}}^{\boldsymbol{\Theta}_l}(t,\eta)\big|_{\beta_{m_i,j}=0} + \bar{r}_{n,\mathrm{inf}}^{\boldsymbol{\Theta}_l}(t,\eta)\big|_{\beta_{m_i,j}\neq 0} \end{aligned} \tag{27}$$

with

$$\begin{aligned} p_{m_0,j}(t) = \Bigg\{ &\left(\boldsymbol{e}^{j}_{m_0,\Omega\{\mathbf{E}_{m_0}(j,:)\}}\right)^T \\ &\times \boldsymbol{s}_l\left(t + \frac{j - \Omega\{\mathbf{E}_{m_0}(j,:)\}}{M_s} T_p\right)\Bigg\} \otimes_t \hat{s}_{m_0}^*(-t) \end{aligned} \tag{28}$$

$$\beta_{m_i,j} = \Omega\{\mathbf{E}_{m_0}(j,:)\} - \Omega\{\mathbf{E}_{m_i}(j,:)\} \tag{29}$$

where $\bar{r}_{n,\mathrm{des}}^{\boldsymbol{\Theta}_l}(t,\eta)$ and $\bar{r}_{n,\mathrm{inf}}^{\boldsymbol{\Theta}_l}(t,\eta)$ denote the match filtering results of the desired signal and the interferences, respectively. It can be observed from (29) that $\beta_{m_i,j} \in [-M_s+1, M_s-1]$ for $m_i \in \boldsymbol{\Theta}_l$ and $j = 1, 2, \ldots, M_s$. As a result, $2M-1$ signal components will be obtained by the matched filtering, with the time delays of the $2M-1$ components being [2]

$$\begin{aligned} \boldsymbol{t_n} = [&t_{n,m_0} - (M_s-1)\Delta\tau, t_{n,m_0} - (M_s-2)\Delta\tau \\ &\ldots, t_{n,m_0} + (M_s-1)\Delta\tau]^T \end{aligned} \tag{30}$$

where t_{n,m_0} denotes the round-trip delay of the desired signal. For a single point-like target, one desired signal and $2(M-1)$ interferences are generated, each having a different location in the range direction [derived from (30)]. Moreover, if the swath is wide enough, $2M-1$ components will occur within one processing cell. As suggested by (30), the incident angles of the $2M-1$ components are different, and thereby, one may suppress the interferences using the DBF processing in the elevation spatial domain. Assuming that the steering vector is a priori known[1], the least-squares (LS) estimate of the weight vector, which will be statistically efficient in the white noise case, can be formed as [36]

$$\begin{aligned} \mathbf{W}_{\mathrm{LS}} &= \mathbf{V}\left(\mathbf{V}^H\mathbf{V}\right)^{-1} \\ &= [\boldsymbol{\omega}_{-M_s+1}, \boldsymbol{\omega}_{-M_s+2}, \ldots, \boldsymbol{\omega}_{M_s-1}] \\ \mathbf{V} &= [\boldsymbol{v}_{-M_s+1}, \boldsymbol{v}_{-M_s+2}, \ldots, \boldsymbol{v}_{M_s-1}] \end{aligned} \tag{31}$$

where $\boldsymbol{v}_0$ and $\boldsymbol{v}_q$, for $q \neq 0$, denote the steering vectors of the desired signal and the interferences, respectively, formed as

$$\begin{aligned} \boldsymbol{v}_q &= \left[e^{j\mu_0(\theta_q)}, e^{j\mu_1(\theta_q)}, \ldots, e^{j\mu_{N_e-1}(\theta_q)}\right]^T \\ \mu_p(\theta) &= 2\pi p d \sin\theta/\lambda, \quad p = 0, 1, \ldots, N_e - 1 \end{aligned} \tag{32}$$

where N_e, d, and λ denote the number of elevation receiving antennas, the channel interval, and the working wavelength, respectively. The set $\boldsymbol{\theta}=[\theta_{-M_s+1}, \theta_{-M_s+2}, \ldots, \theta_{M_s-1}]$ contains the DOAs of the desired signal (θ_0) and the interferences (θ_q, for $q \neq 0$). For the instantaneous slant range r_0

$$\theta_q = \arcsin\left\{H_0 \Big/ \left(r_0 + \frac{c}{2} \cdot \frac{qT_p}{M_s}\right)\right\}. \tag{33}$$

As indicated by (31), to form an accurate estimate of the weight vector, the matrix $\mathbf{V}^H\mathbf{V} \in \mathbb{C}^{2M_s-1 \times 2M_s-1}$ must be invertible, i.e.,

$$\begin{aligned} \mathrm{rank}\left(\mathbf{V}^H\mathbf{V}\right) &= \mathrm{rank}\left(\mathbf{V}\right) \\ &= \min\left(N_e, 2M_s - 1\right) = 2M_s - 1. \end{aligned} \tag{34}$$

It can be concluded from (34) that, for $\mathbf{V}^H\mathbf{V}$ to be invertible, we should satisfy $N_e \geq 2M_s - 1$. That is, to achieve the echo separation by the current STSO scheme, at least $2M_s-1$ receiving antennas in the elevation are required for the M_s transmitted waveforms. It is worth remarking that the use of sufficient DOFs is also crucial for the acquirement of satisfactory image quality. Therefore, for DOF-limited remote sensing systems, this can pose a considerable challenge in terms of the array configuration. In order to alleviate this problem, we here propose a novel separation solution. This solution can be divided into two main parts: spectrum preprocessing and segment synthesis.

1) Spectrum Preprocessing: In order to reduce the system resources required for DBF processing, the number of interference components in the cell under processing must be reduced. To do so, the bandpass filter bank is here instead implemented in the range frequency domain. For M_s transmitting waveforms, the fast time spectrum is divided into M_s subbands. To filter out the echoes in the jth range frequency subband, we define the bandpass filters as

$$\begin{aligned} &w_j^{r,\mathrm{BF}}(f_t, \eta) \\ &= \mathrm{rect}\left(\frac{f_t - \frac{\xi\left(\frac{(j-1)T_p}{M_s} - \frac{T_p}{2}\right) + \xi\left(\frac{jT_p}{M_s} - \frac{T_p}{2}\right)}{2}}{\left|\xi\left(\frac{(j-1)T_p}{M_s} - \frac{T_p}{2}\right) - \xi\left(\frac{jT_p}{M_s} - \frac{T_p}{2}\right)\right|}\right) \end{aligned} \tag{35}$$

[1]In the alternative case, i.e., when the steering direction is not a priori known, the directions of arrival (DOAs) of the signal components at each instant of the recorded time-window may be estimated using various approaches, e.g., Capon or MUSIC [19].

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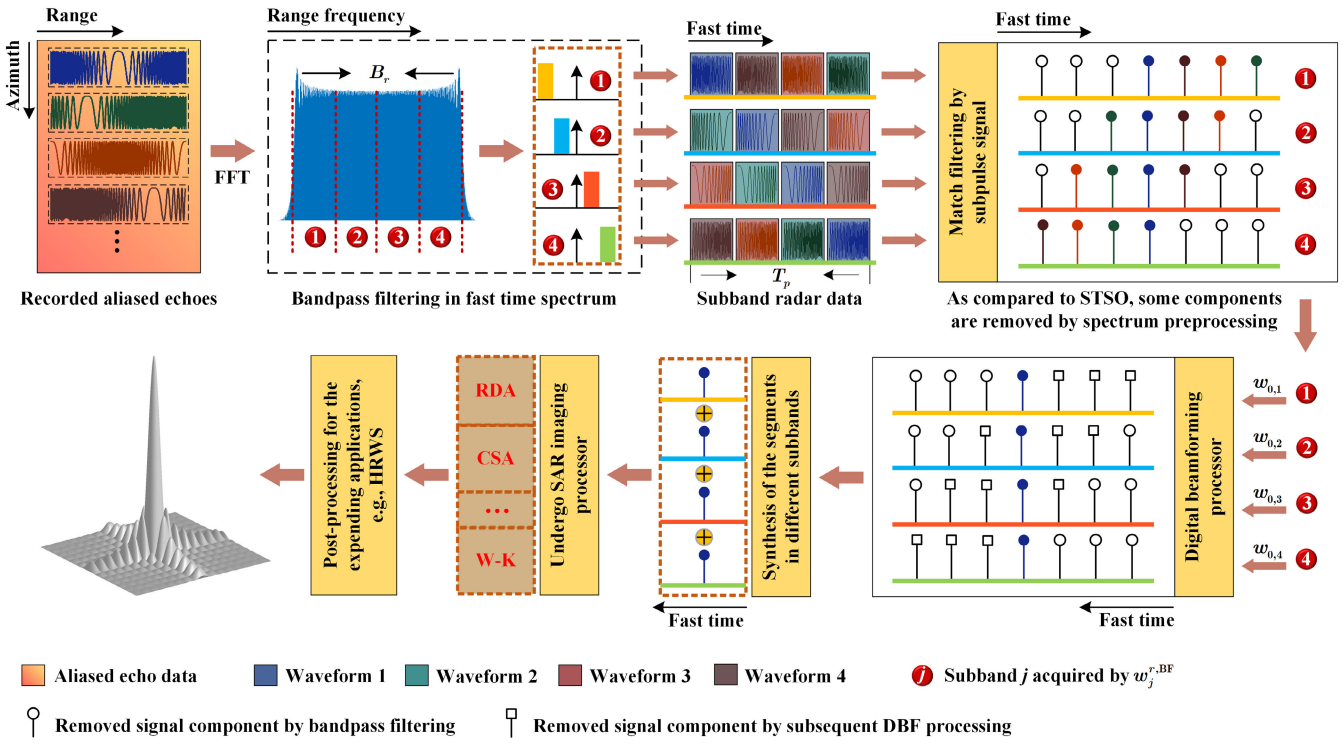


Fig. 5. Flowchart of spectrum preprocessing and segment synthesis. Four waveforms, as expressed in (10) and (12), are transmitted by the MIMO-SAR systems. Only the separation of waveform 1 (i.e., $\bar{s}_l(t)$) is shown in this figure. In addition, it is worth mentioning that, in the DBF processing, the optimal weight vectors $\omega_{0,j}$ for multiple subbands are different.

with

$$-\frac{B_r}{2} \le \xi(t) \le \frac{B_r}{2}, \quad -\frac{T_p}{2} \le t \le \frac{T_p}{2} \tag{36}$$

where B_r and $\xi(t)$ are the bandwidth and the time–frequency function of $\bar{s}_l(t)$ in (9), respectively, as shown in Fig. 4. Filtering $r_n^{\mathbf{\Theta}_l}(t,\eta)$ through the bandpass filter in (35) yields

$$\begin{aligned}\tilde{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta) &= \mathcal{F}_t^{-1}\left\{\mathcal{F}_t\left(\tilde{r}_n^{\mathbf{\Theta}_l}(t,\eta)\right) w_j^{\mathrm{r,BF}}(f_t,\eta)\right\} \\ &= \sum_{m_i\in\mathbf{\Theta}_l} w_j\left(t - t_{n,m_i} + \frac{j - \Omega\{\mathbf{E}_{m_i}(j,:)\}}{M_s}T_p\right) \\ &\quad \times \tilde{r}_{n,m_i}(t,\eta)\end{aligned} \tag{37}$$

where $w_j(t)$ denotes the window function in (1). On proceeding, $\tilde{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta)$ is match filtered by $\hat{s}_{m_0}(t)$, yielding

$$\begin{aligned}\bar{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta) &= \tilde{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta) \otimes_t \hat{s}_{m_0}^*(-t) \\ &= \sum_{m_i\in\mathbf{\Theta}_l} h_{n,m_i}(t,\eta) \otimes_t p_{m_0,j}\left(t + \frac{\beta_{m_i,j}}{M_s}T_p\right).\end{aligned} \tag{38}$$

According to (29), the set of $\beta_{m_i,j}$ is inconsistent for different range frequency subbands although the set always contains M_s elements. It is worth noting that, for the signal segment $\bar{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta)$, only the aliased echoes in subband j are match filtered. Therefore, the aliased echoes are match filtered by the subpulse signal instead of $\hat{s}_{m_0}(t)$. As a result, (38) may be written as

$$\begin{aligned}\bar{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta) &= \tilde{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta) \otimes_t \hat{s}_{m_0,j}^*(-t) \\ &= \sum_{m_i\in\mathbf{\Theta}_l} h_{n,m_i}(t,\eta) \otimes_t p'_{m_0,j}\left(t + \frac{\beta_{m_i,j}}{M_s}T_p\right) \\ &= \bar{\tilde{r}}_{n,\mathrm{des}}^{\mathbf{\Theta}_l,j}(t,\eta)\big|_{\beta_{m_i,j}=0} + \bar{\tilde{r}}_{n,\mathrm{inf}}^{\mathbf{\Theta}_l,j}(t,\eta)\big|_{\beta_{m_i,j}\neq 0}\end{aligned} \tag{39}$$

Algorithm 1 Proposed DBF Processing Solution

Input : The aliased signal returns $\tilde{r}_{n,m}(t,\eta)$ which occupy the Doppler subband l before Doppler deviation correction;

Output: The echo separation result $r_{n,m}^{\mathrm{separated}}(t,\eta)$ radiated by Txm and received by Rxn;

for $j = 1 : M_s$ **do**
- Bandpass filtering $\tilde{r}_{n,m}(t,\eta)$ by $w_j^{r,\mathrm{BF}}$, causing the spectrum of $\tilde{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta)$ to occupy $\xi((j-1)T_p/M_s - T_p/2) \sim \xi(jT_p/M_s - T_p/2)$;
- Match filtering $\tilde{\tilde{r}}_n^{\mathbf{\Theta}_l,j}(t,\eta)$ by the subpulse signal $\hat{s}_{m_0,j}(t)$;
- Using the steering vector matrix $\mathbf{V}_j$, calculate the LS weight vector $\boldsymbol{\omega}_{0,j}$ to suppress the interferences in subband j;

end

Synthesis of desired signal segments using (44).

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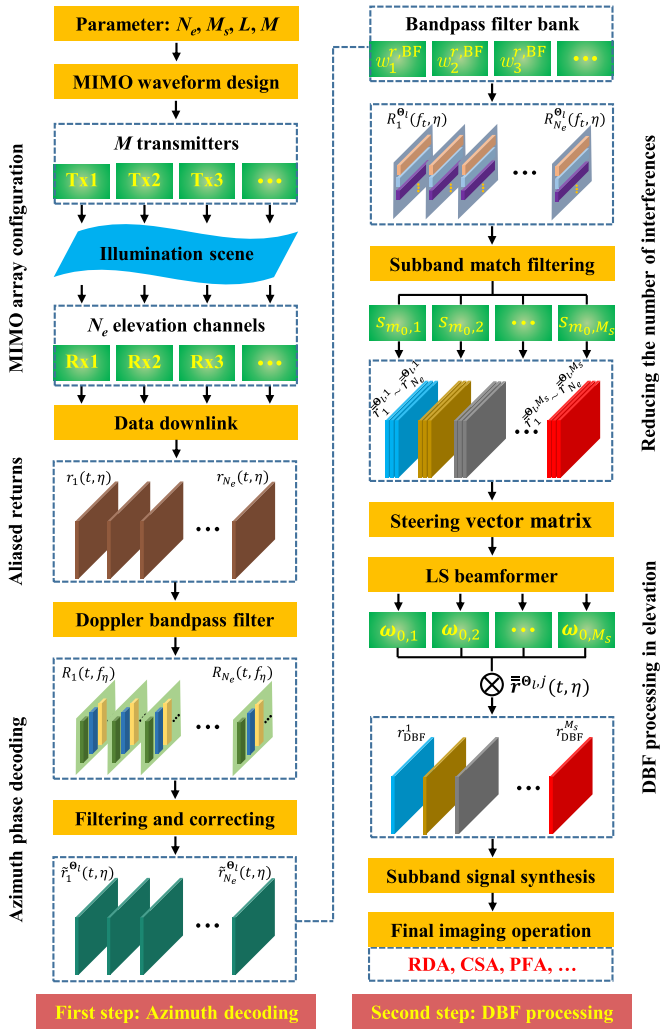


Fig. 6. Echo separation procedures of the proposed method. The proposed generic MIMO-SAR solution contains the following parts: array configuration and waveform design, azimuth phase decoding in the Doppler domain, DBF processing in the elevation spatial domain (spectrum preprocessing and segment synthesis), and the final SAR imaging processing.

where $\hat{s}_{m_0,j}(t)$ and $p'_{m_0,j}(t)$ represent the subpulse signal in the jth subband and the autocorrelation of $\hat{s}_{m_0,j}(t)$, respectively. In addition, $\hat{s}_{m_0,j}(t)$ can be expressed as

$$\hat{s}_{m_0,j}(t) = \left\{ \left(\boldsymbol{e}^{j}_{m_0,\Omega\{\mathbf{E}_{m_0}(j,:)\}} \right)^{T} \times \boldsymbol{s}_l \left(t + \frac{j - \Omega\{\mathbf{E}_{m_0}(j,:)\}}{M_s} T_p \right) \right\}. \quad (40)$$

By combining (27) and (39), one may conclude that the number of interferences has been halved due to the spectrum preprocessing, which can reduce the required system resources in the process of DBF on receive in elevation.

2) Segment Synthesis: Next, focusing on the suppression of interference segments in $\bar{\bar{r}}_n^{\boldsymbol{\Theta}_l,j}(t,\eta)$, i.e., $\bar{\bar{r}}_{n,\mathrm{inf}}^{\boldsymbol{\Theta}_l,j}(t,\eta)$, a weight vector formed similar to (31) is employed. Note that, due to the reduced number of interference segments, the MIMO-SAR systems only need to satisfy $N_e \geq M_s$. Based on (29), the corresponding steering matrix of the jth range frequency subband is defined as

$$\mathbf{V}_j = [\boldsymbol{v}_{q\prime} | q\prime = \Omega\{\mathbf{E}_{m_0}(j,:)\} - \Omega\{\mathbf{E}_{m_i}(j,:)\} \; m_i \in \boldsymbol{\Theta}_l, \Omega\{\mathbf{E}_{m_i}(j,:)\} = 1,2,\ldots,M_s]. \quad (41)$$

Substituting (41) into (31), one obtains the LS weight vector $\boldsymbol{\omega}_{0,j}$, i.e.,

$$\mathbf{W}_{\mathrm{LS}}^{j} = \mathbf{V}_j \left(\mathbf{V}_j^H \mathbf{V}_j \right)^{-1} = [\boldsymbol{\omega}_{q\prime,j} | q\prime = \Omega\{\mathbf{E}_{m_0}(j,:)\} - \Omega\{\mathbf{E}_{m_i}(j,:)\}]. \quad (42)$$

Generally, the discretized match filtered outputs, $\bar{\bar{r}}_n^{\boldsymbol{\Theta}_l,j}(t,\eta)$, are organized in the same manner as the steering vector $\boldsymbol{v}_{q'}$ in (41) to acquire length-N_e spatial snapshots. Thus, the DBF processing result of the jth subband is

$$r_{\mathrm{DBF}}^{j}(t,\eta) = \boldsymbol{\omega}_{0,j}^{H} \bar{\bar{\boldsymbol{r}}}^{\boldsymbol{\Theta}_l,j}(t,\eta) = \boldsymbol{\omega}_{0,j}^{H} \left[\bar{\bar{r}}_1^{\boldsymbol{\Theta}_l,j}(t,\eta), \bar{\bar{r}}_2^{\boldsymbol{\Theta}_l,j}(t,\eta), \ldots, \bar{\bar{r}}_{N_e}^{\boldsymbol{\Theta}_l,j}(t,\eta) \right]^{T}. \quad (43)$$

To proceed further, the echoes of $\hat{s}_{m_0}(t)$ in the different range frequency subbands should be separated to reconstruct the radar signal radiated by the m_0th Tx and received by Rxn. According to (43), the separated single channel response can be expressed as

$$r_{n,m}^{\mathrm{separated}}(t,\eta) = \sum_{j=1}^{M_s} \boldsymbol{\omega}_{0,j}^{H} \bar{\bar{\boldsymbol{r}}}^{\boldsymbol{\Theta}_l,j}(t,\eta). \quad (44)$$

The step-by-step DBF processing procedure is summarized in Algorithm 1, with Fig. 5 showing the corresponding flowchart for the case of M_s=4. Using the spectrum preprocessing and segment synthesis steps, the proposed method can reduce the system cost of processing the echo separation, which, thereby, enables the production of low-cost and low-mass SAR sensors. The proposed generic MIMO-SAR solution scheme is outlined in Fig. 6.

IV. Discussion

We proceed to examine some of the tradeoffs and practical considerations that have to be taken into account. It is anticipated that the reconfigurable waveform transmission framework and echo separation solution may be leveraged to further improve the MIMO-SAR implementations.

A. Range Resolution

In this section, we examine the effects of the bandpass filtering (in fast time spectrum) and the segment synthesis on the range resolution[2]. Generally, for the current STSO beamforming scheme, the match filtering result of $\hat{s}_{m_0}(t)$ can be expressed as [see (27)–(29)]

$$\bar{r}_{n,m_0}(t,\eta) = \tilde{r}_{n,m_0}(t,\eta) \otimes_t \hat{s}^{*}_{m_0}(-t) = \sum_{j=1}^{M_s} h_{n,m_0}(t,\eta) \otimes_t p_{m_0,j}(t) = \bar{r}_{n,\mathrm{des}}^{\boldsymbol{\Theta}_l}(t,\eta). \quad (45)$$

[2]The bandpass filtering and segment synthesis are performed in the range direction, and thereby, they have no effect on the azimuth resolution.

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Applying the weight vector calculated in (31) yields

$$r_{n,m_0}^{\text{separated}}(t,\eta)=\boldsymbol{\omega}_0^H\overline{\boldsymbol{r}}^{m_0}(t,\eta)=\boldsymbol{\omega}_0^H\boldsymbol{v}_0\overline{r}_{1,m_0}(t,\eta)$$
$$=\boldsymbol{\omega}_0^H\boldsymbol{v}_0\sum_{j=1}^{M_s}h_{1,m_0}(t,\eta)\otimes_t p_{m_0,j}(t) \tag{46}$$

$$\overline{\boldsymbol{r}}^{m_0}(t,\eta)=\left[\overline{r}_{1,m_0}(t,\eta),\overline{r}_{2,m_0}(t,\eta),\ldots,\overline{r}_{N_e,m_0}(t,\eta)\right]. \tag{47}$$

Generally, under the far-field assumption, the effect that the time-delay difference between different Txs has on $\delta(t)$ can be neglected in (46). In the proposed separation solution, the range compression result of range frequency subband j is expressed as [see (39)]

$$\begin{aligned}\bar{\bar{r}}_n^{m_0\in\boldsymbol{\Theta}_l,j}(t,\eta)&=\tilde{\bar{r}}_n^{m_0\in\boldsymbol{\Theta}_l,j}(t,\eta)\otimes_t\hat{s}_{m_0,j}^*(-t)\\&=h_{n,m_0}(t,\eta)\otimes_t p'_{m_0,j}(t)\\&=\bar{\bar{r}}_{n,\text{des}}^{\boldsymbol{\Theta}_l,j}(t,\eta)\end{aligned} \tag{48}$$

where $\tilde{\bar{r}}_n^{m_0\in\boldsymbol{\Theta}_l,j}(t,\eta)$ represents the subband echoes of $\hat{s}_{m_0}(t)$. Without loss of generality, only the recorded signal transmitted by m_0 Tx is match filtered, as described in (48). One then performs the DBF processing and the segment synthesis on $\bar{\bar{r}}_n^{m_0\in\boldsymbol{\Theta}_l,j}(t,\eta)$, yielding the separated MIMO-SAR data [see (43) and (44)]

$$\begin{aligned}r_{n,m}^{\text{separated}}(t,\eta)&=\sum_{j=1}^{M_s}\boldsymbol{\omega}_{0,j}^H\bar{\bar{\boldsymbol{r}}}^{m_0\in\boldsymbol{\Theta}_l,j}(t,\eta)\\&=\sum_{j=1}^{M_s}\boldsymbol{\omega}_{0,j}^H\boldsymbol{v}_0\left\{h_{1,m_0}(t,\eta)\otimes_t p'_{m_0,j}(t)\right\}.\end{aligned} \tag{49}$$

It should be stressed that the beamformer can generate a distortionless response to the desired signal [37] such that

$$\begin{aligned}&\boldsymbol{\omega}_0^H\boldsymbol{v}_0=1\\&\boldsymbol{\omega}_{0,1}^H\boldsymbol{v}_0=\boldsymbol{\omega}_{0,2}^H\boldsymbol{v}_0=\cdots=\boldsymbol{\omega}_{0,M_s}^H\boldsymbol{v}_0=1.\end{aligned} \tag{50}$$

Substituting (50) into (49) yields

$$\begin{aligned}r_{n,m}^{\text{separated}}(t,\eta)&=\sum_{j=1}^{M_s}h_{1,m_0}(t,\eta)\otimes_t p'_{m_0,j}(t)\\&=\sum_{j=1}^{M_s}h_{1,m_0}(t,\eta)\otimes_t p_{m_0,j}(t).\end{aligned} \tag{51}$$

Consequently, one can conclude that (46) and (51) have the same form. Therefore, the spectrum preprocessing and the segment synthesis in the proposed method will have no effect on the range resolution.

B. System Cost in Echo Separation

In the proposed solution, the separation of the aliased echoes is achieved by interpulse decoding in azimuth and DBF processing in the elevation domain. The interpulse decoding technique requires that the azimuth sampling frequency satisfies $F_a\geq LB_D$, which, in turn, reduces the swath width and requires a higher downlink capacity. To reduce the requirement on the PRF, azimuth multichannel techniques can be employed to reduce the spectrum aliasing caused by APC. In this case, the PRF only needs to be slightly larger than B_D, but the echo separation is achieved at the cost of azimuth DOFs [29]. Generally, the azimuth DOFs are of notable importance for SAR imaging performance, especially for HRWS imaging. Considering the abovementioned problems, the system resources in elevation are fully exploited by the reconfigurable waveform transmitting framework shown in Fig. 2.

Moreover, to separate the aliased signal returns in Doppler subband l (i.e., the echoes of the M_s STSO waveforms), additional DBF in elevation is required for a wide swath. For beamformers using a priori knowledge of the signal segments, such as LS [36] or linear constrained minimum variance (LCMV) [19], the MIMO-SAR systems should use M_s elevation subapertures in the proposed solution. However, for the robust beamformers, such as the derivative-constraint-based [17] or Capon spatial spectrum integral-based beamformers [38], [39], the number of subapertures must satisfy $N_e{>}M_s$ since the beamformers acquire the robustness with the cost of system DOFs[3].

In summary, to generate better echo separation performance, we should comprehensively consider the system resources in both the Doppler and elevation spatial domains. For example, for the reconfigurable MIMO-SAR systems with $M=4$, one may use

$$\begin{cases}L=2,M_s=2,F_a\geq 2B_D,N_e\geq 2\\L=1,M_s=4,B_D<F_a<2B_D,N_e\geq 4.\end{cases} \tag{52}$$

In the proposed solution, the system resources in the elevation spatial domain are preferentially adopted to achieve the separation of aliased signal returns.

C. Computational Complexity

We proceed to compare the computational complexity of the DBF processing in the proposed method with that of the current STSO beamforming solution. To separate the aliased radar data in the range direction, the proposed method can be divided into three steps: 1) subband division by fast time spectrum preprocessing; 2) match filtering and DBF processing for each range frequency subband; and 3) segment synthesis. The second step implies that a cell under processing undergoes the DBF processing M_s times, whereas, in the current STSO beamforming framework, it is processed only once. As a result, the computational complexity of the proposed method is M_s times that of the current STSO scheme (in the worst cases).

The computational complexity for the LS beamformer is listed in Table I, where N_e and N_e' denote the number of elevation antennas used by the two schemes, respectively. In addition, due to the required M_s DBF processing steps, the total computational complexity of the proposed method is $\mathcal{O}(M_s^3(2N_e'{+}M_s))$. In order to achieve the lowest computational load, one may use $N_e{=}2M_s{-}1$ and $N_e'{=}M_s$. In such a

[3]For the Capon spatial spectrum integral beamformer and the ESB beamformer, $N_e{\geq}M_s{+}1$. For the derivative-constraint-based beamformer, $N_e{\geq}(P{+}1)M_s$ should be met to widen the mainlobe and the nulls, where P represents the derivative order.

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TABLE I
COMPUTATIONAL COMPLEXITY FOR THE LS BEAMFORMER

Methods	Matrix calculation ($\mathbf{V}^{\mathrm{H}}\mathbf{V}$)	Matrix inversion	Weight vector calculation	Sum
Current STSO scheme	$\mathcal{O}((2M_s-1)^2N_e)$	$\mathcal{O}((2M_s-1)^3)$	$\mathcal{O}((2M_s-1)^2N_e)$	$\mathcal{O}((2M_s-1)^2(2N_e+2M_s-1))$
Proposed method	$\mathcal{O}(M_s^2N_e')$	$\mathcal{O}(M_s^3)$	$\mathcal{O}(M_s^2N_e')$	$\mathcal{O}(M_s^2(2N_e'+M_s))$

TABLE II
COMPARISON IN TERMS OF THE EXECUTION TIME

Value of M_s	$M_s=2$	$M_s=3$	$M_s=4$	$M_s=5$
STSO scheme	43.33us	48.35us	54.16us	64.05us
Proposed scheme	42.57us	45.39us	49.57us	59.98us

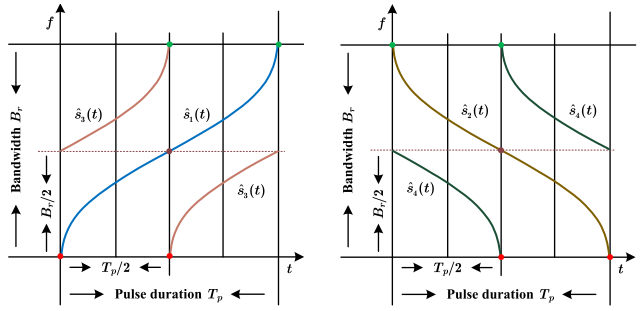


Fig. 7. Time–frequency structures for (a) waveforms $\hat{s}_1(t)$ and $\hat{s}_3(t)$ and (b) waveforms $\hat{s}_2(t)$ and $\hat{s}_4(t)$.

case, the computational complexities of the standard and proposed schemes are $\mathcal{O}(3(2M_s-1)^3)$ and $\mathcal{O}(3M_s^4)$, respectively. Comparing these

$$\mathcal{O}\left(3\left(2M_s-1\right)^3\right) > \mathcal{O}\left(3M_s^4\right), \quad M_s = 2,3,\ldots,6 \tag{53}$$

indicating that the computational complexity of the proposed method is lower than that of the current STSO scheme in some cases. This is also illustrated in Table II, showing the execution time[4] for $N_e{=}2M_s{-}1$ and $N_e'{=}M_s$. As can be seen in the table, the execution time of the proposed method, for different M_s, is slightly less than that of the current STSO scheme. As a result, the running time is not considered to be a limiting factor for the proposed scheme even though the array configuration has been significantly simplified.

V. SIMULATION AND EXPERIMENT ANALYSIS

In this section, the feasibility and performance of the proposed method in terms of echo separation are investigated for a spaceborne MIMO-SAR system, comparing the obtained results with that of the current STSO beamforming solution. These simulations can be divided into two parts: 1) point-like simulations and 2) distributed scene simulations. In these simulations, two waveform design schemes based on Fig. 2 are considered to radiate four NLFM waveforms, i.e.,

$$\begin{aligned} &\text{case 1}: \ L=2, \quad M_s=2 \\ &\text{case 2}: \ L=1, \quad M_s=4. \end{aligned} \tag{54}$$

[4]The execution time of the two methods is calculated by the same simulation tool and indicates the required time of the DBF operation on a single cell under processing. It should be noted that the M_s DBF operations required by the proposed method are here carried out in parallel.

TABLE III
SPACEBORNE MIMO-SAR SYSTEM PARAMETERS

Parameter	Value
Bandwidth	60 MHz
Pulse duration	80 us
Center frequency	1.26 GHz
PRF	3320 Hz
Wavelength	0.238 m
Array type	Planar
Number of subapertures in elevation	4
Number of subapertures in azimuth	1
Antenna height in elevation	5 m
Antenna length in azimuth	10 m
Radar altitude	607 km
Center of illumination scene	670 km
Radar velocity	7553 m/s
Look angle access range	23.2°∼26.8°

The considered radar system works in the stripmap mode, with the major system parameters (see [7] for the parameters of the LT-1 spaceborne SAR system) being listed in Table III. In addition, the beam patterns of the Tx and Rx are assumed to have the same form and are generated by a raised cosine function [3].

A. Point-Like Target Simulation

To provide insights into the performance of the proposed method, we initially examine detailed point-like target simulations. We employ the case 1 transmitting scheme with the radiated waveforms being designed using (10). The four waveforms are divided into two sets, which are both modulated by the APC techniques. Fig. 7 illustrates the time–frequency structures for the two sets of waveforms. Here, the basic NLFM signals $\overline{s}_1(t)$ and $\overline{s}_2(t)$ are generated by a Taylor window with a maximum sidelobe level of -40 dB.

As shown in Fig. 8, four point-like targets located at different range and azimuth cells are illuminated by $\hat{s}_1(t)$, $\hat{s}_2(t)$, $\hat{s}_3(t)$, and $\hat{s}_4(t)$. To achieve the echo separation, azimuth phase decoding is implemented on the aliased signal returns. Based on the bandpass filtering in the Doppler domain, one may acquire two sets of signal returns: 1) the aliased signals transmitted by $\hat{s}_1(t)$ and $\hat{s}_3(t)$ and 2) the signals radiated by $\hat{s}_2(t)$ and $\hat{s}_4(t)$.

Taking the first set of aliased signal returns as an illustrative example, to further separate $\hat{s}_1(t)$ and $\hat{s}_3(t)$, one needs to implement the DBF processing in the elevation spatial domain in order to retrieve the echoes of interest while suppressing the interferences. The improved imaging performance provided by the proposed method can be seen by comparing the proposed solution with the current STSO and SPC schemes. Fig. 9 shows the echo signals and the point spread functions (PSFs)

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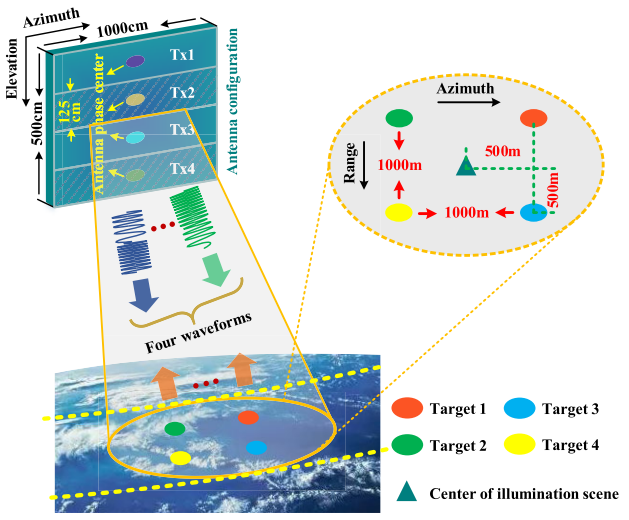


Fig. 8. Illustration of the point target imaging scene.

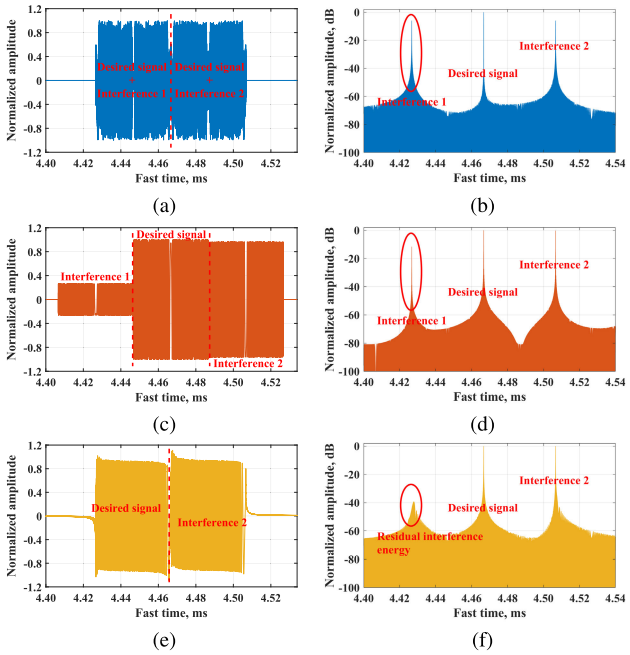


Fig. 9. Comparison between the current STSO scheme, SPC scheme, and the proposed method. Point target echoes of (a) STSO scheme, (c) SPC scheme, and (e) proposed method. Match filtering results by $\hat{s}_1(t)$ for (b) STSO scheme, (d) SPC scheme, and (f) proposed method. The figure in (f) indicates the match filtering result of the first subband occupying $-B_r/2\sim0$.

achieved by the mentioned three schemes. As may be seen in the figure, there are two interference components in the current STSO and SPC schemes, while the proposed method has notably reduced the influence of interference 1. However, due to the energy leakage[5] from other subbands, there still remains some interference energy at the location of interference 1. Considering that the energy of the residual interference is substantially lower (about 40 dB lower than that of the desired signal), the effects of the residual interference may typically be neglected in the following analysis. Due to the reduced number of interference segments, DBF processing can be achieved with a relatively low cost of system DOFs, illustrating the superiority of the proposed method.

Finally, the range-Doppler algorithm (RDA) is employed for the imaging processing of the separated echoes (particularly, the aliased returns are separated through the beamforming of the dual-channel radar data). The target responses acquired by the proposed method are shown in Fig. 10. The impulse response width (IRW), the peak sidelobe ratio (PSLR), and the integrated sidelobe ratio (ISLR) [31] are calculated to quantitatively evaluate the imaging performance. It is evident that all the point-like targets are well-focused, indicating the proposed method's effectiveness in terms of echo separation. Moreover, with the employment of the NLFM signal, the sidelobe in the range direction has been suppressed substantially.

B. Distributed Scene Simulation

In order to further illustrate the reliability of the proposed method, we proceed to examine simulations of a distributed scenario. Here, we use the two transmit waveforms in (54) (with the same array configuration) for the distributed scene. The relevant simulation parameters are listed in Table III.

1) Case 1: To guarantee that the image information is covered by the interference segments (i.e., additional beamforming is required for echo separation), the delay length t_W of the swath must be longer than $T_p/2$. However, to highlight the superiority of the spectrum preprocessing and the segment synthesis, the delay length of the simulated swath is here longer than T_p (i.e., three signal components, at most, will exist in a cell under processing).

The separation result of $\hat{s}_3(t)$ is shown in Fig. 11. Fig. 11(a) illustrates the original scenario illuminated by $\hat{s}_3(t)$, i.e., the ground truth. It should be stressed that the waveform orthogonality cannot be ensured if $t_W \geq T_p/2$ and the induced interference energy can seriously deteriorate the image quality [as illustrated in Fig. 11(b)]. Subsequently, the aliased signal returns of $\hat{s}_1(t)$ and $\hat{s}_3(t)$ are separated by the current STSO scheme and the proposed method with the same system resources (i.e., two receiving antennas in the elevation). It can be observed that the interference energy in Patch 2 and Patch 3 can be suppressed by all schemes with similar resulting image quality. However, in Fig. 11(c) and (d), the separation performance of the current STSO and SPC schemes in Patch 1 can be seen to be seriously deteriorated due to that the system DOFs are not enough to form a beamformer that can produce sufficient nulls in the directions of the interferences though the desired mainlobe echoes are extracted without signal loss. Based on the spectrum preprocessing, the number of interference segments has been halved, which suggests that the system resources are adequate for echo separation. As shown in Fig. 11(e), the proposed method can also be seen to acquire better image quality, being close to the ground truth.

To quantitatively evaluate the echo separation performance in Fig. 11, the image metrics (i.e., contrast and entropy)[6] [39], [40] are utilized. The results for different patches are

[5]Generally, the subband j acquired by the bandpass filter also contains the echoes from the adjacent subbands, which may result in additional signal components in the filtered signals. Essentially, additional components are induced by the discontinuous time–frequency curve (see Fig. 7).

[6]A higher contrast or a lower entropy indicates a better image quality.

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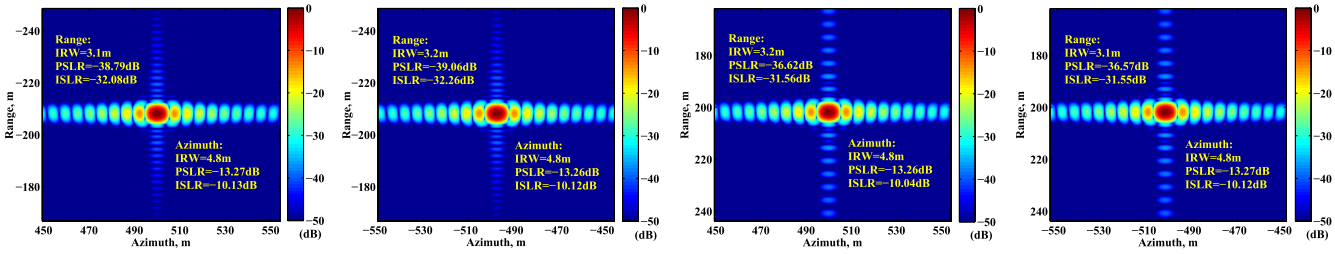


Fig. 10. Target responses acquired by the proposed method: (a) target 1, (b) target 2, (c) target 3, and (d) target 4. The targets are illuminated by $\hat{s}_1(t)$, $\hat{s}_2(t)$, $\hat{s}_3(t)$, and $\hat{s}_4(t)$, respectively. Here, the Y-axis denotes the range referenced to the closest slant range R_0.

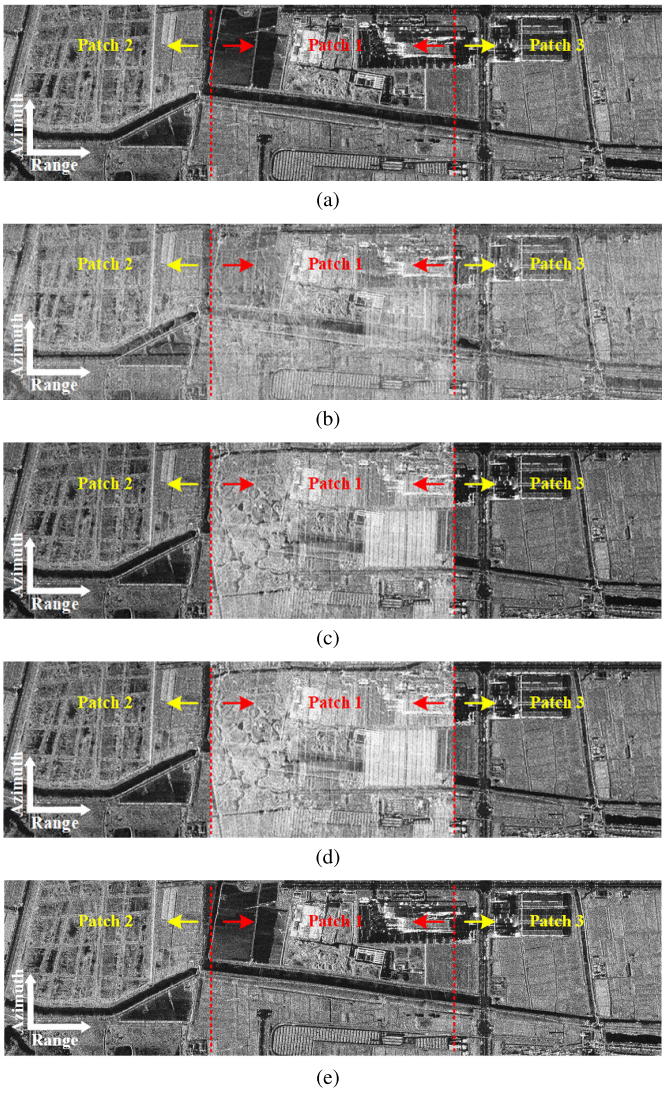


Fig. 11. Distributed scene processing results in case 1. (a) Ground truth. Imaging results of (b) aliased returns (radiated by $\hat{s}_1(t)$ and $\hat{s}_3(t)$), (c) STSO scheme, (d) SPC scheme, and (e) proposed method. Here, the image size is 12 km (azimuth) × 45 km (range) to guarantee $t_W \geq T_p$.

TABLE IV
CONTRAST AND ENTROPY FOR THE THREE SCHEMES

		Fig. 11(a)	Fig. 11(b)	Fig. 11(c)	Fig. 11(d)	Fig. 11(e)
Patch 1	Contrast	2.604	1.511	1.673	1.675	2.600
	Entropy	15.426	16.007	15.964	15.960	15.429
Patch 2	Contrast	2.016	1.621	2.015	2.015	2.015
	Entropy	15.572	15.815	15.573	15.572	15.572
Patch 3	Contrast	2.248	1.779	2.245	2.244	2.245
	Entropy	15.472	15.759	15.475	15.475	15.473

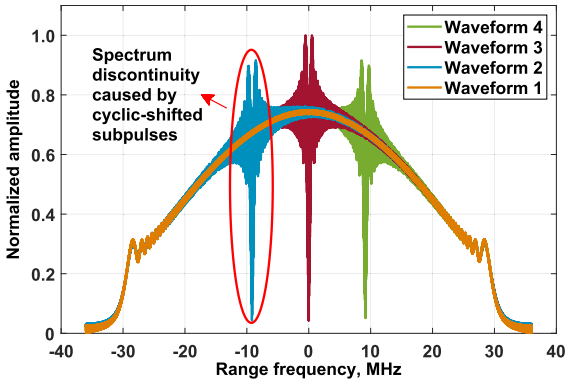


Fig. 12. Spectra of the radiated waveforms. Since waveforms 2–4 are composed of cyclic-shifted subpulses, a spectrum discontinuity occurs in the corresponding spectra.

summarized in Table IV. As expected, the image metrics are consistent with the earlier analysis, which demonstrates the effectiveness of the proposed method.

2) Case 2: In most practical cases, the azimuth sampling frequency is relatively low in order to avoid range ambiguity and reduce the burden of the data downlink, which can lead to limited resources in the Doppler domain. In this respect, another waveform transmitting scheme [i.e., case 2 in (54)] is adopted to verify the reliability of the proposed method. In this situation, all the elevation subapertures are utilized to implement the DBF on receive. Fig. 12 illustrates the resulting fast time spectra of the four radiated waveforms.

To simplify comparisons, the simulated distributed scene is the same as that shown in Fig. 11(a). The recorded aliased signal returns are match filtered by $\hat{s}_3(t)$, with the resulting imaging result being shown in Fig. 13(b). It can be observed that the image information is submerged in the widespread interference energy. According to the four-channel MIMO-SAR radar data, the interferences in Patch 1, Patch 2, and Patch 3 cannot be completely suppressed by STSO and SPC scheme (only three interference components can be removed by the resulting nulls). However, the proposed method can halve the number of interference segments by the use of

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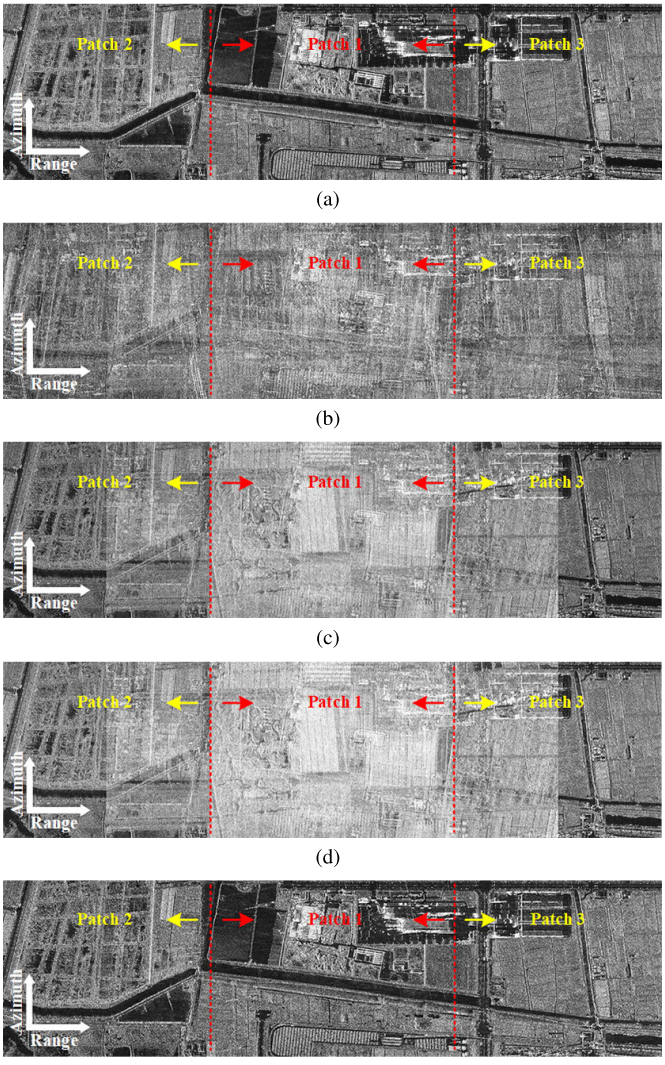


Fig. 13. Distributed scene processing results in case 2. (a) Ground truth. Imaging results of (b) aliased signal returns, (c) current STSO scheme, (d) SPC scheme, and (e) proposed method.

TABLE V
CONTRAST AND ENTROPY FOR THE THREE SCHEMES

		Fig. 13(a)	Fig. 13(b)	Fig. 13(c)	Fig. 13(d)	Fig. 13(e)
Patch 1	Contrast	2.604	1.310	1.369	1.368	2.438
	Entropy	15.426	16.124	16.080	16.080	15.550
Patch 2	Contrast	2.016	1.316	1.785	1.785	1.907
	Entropy	15.572	15.960	15.694	15.694	15.647
Patch 3	Contrast	2.248	1.337	1.852	1.853	2.061
	Entropy	15.472	15.963	15.670	15.670	15.600

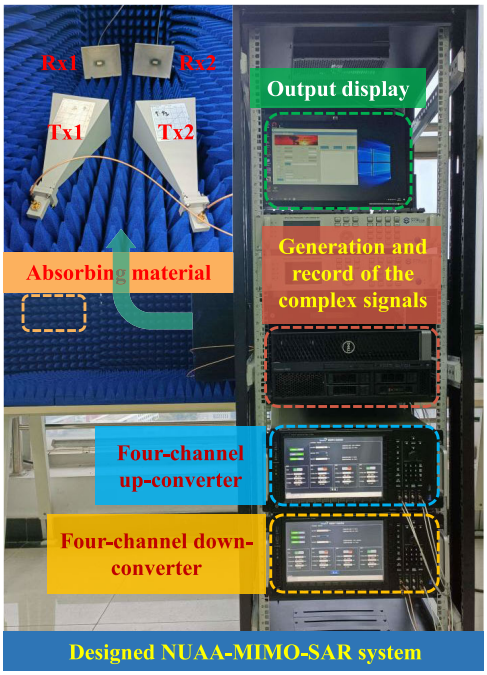


Fig. 14. Diagram of the system equipment to be deployed on the airborne platform.

spectrum preprocessing. In this respect, the system resources provided by the MIMO-SAR system are enough to implement the beamforming in elevation. The final imaging result of the proposed method is shown in Fig. 13(e). To further demonstrate the performance of the proposed method, the contrast and entropy of Fig. 13 are listed in Table V. It can be again seen in both Fig. 13 and Table V that the proposed method is preferable to the standard approach.

C. Ground Experiment Analysis

In the process of developing an airborne MIMO-SAR system, many practical imperfections and implementation issues, such as phase, gain, and offset mismatch of the analog-to-digital converter (ADC) or the digital-to-analog converter (DAC), must be considered. As shown in Fig. 14, the considered MIMO system can be divided into several parts, such as the generation and recording of the complex electromagnetic wave signals, upconverter, transmit/receive antenna, and downconverter. The main device metrics for the system are listed in Table VI. In our future work, flight experiments will be performed to verify the performance of the designed MIMO-SAR system. For now, only ground-based experiments have been carried out to verify the Tx/Rx data synchronization performance[7] and the system coherent performance. The experiment parameters are listed in Table VII.

Fig. 15 shows the processing results of the measured data, as well as the comparison between the simulations and the corresponding measurements. Fig. 15(a) and (b) indicates the range spectra of $\hat{s}_1(t)$ and $\hat{s}_2(t)$, respectively. Compared to the ideal signal spectra, the spectra resulting from the measurement can be seen to have the same signal bandwidth and a similar spectral shape though some interference can also be seen to exist in electromagnetic wave propagation. Subsequently, the recorded data undergo a match filter, with the processing result being displayed in Fig. 15(c). From the results in Fig. 9, it may be expected that one interference (marked by ellipse) would be possible to remove by the

[7]This can be divided into two parts: transmitting synchronization and receiving synchronization.

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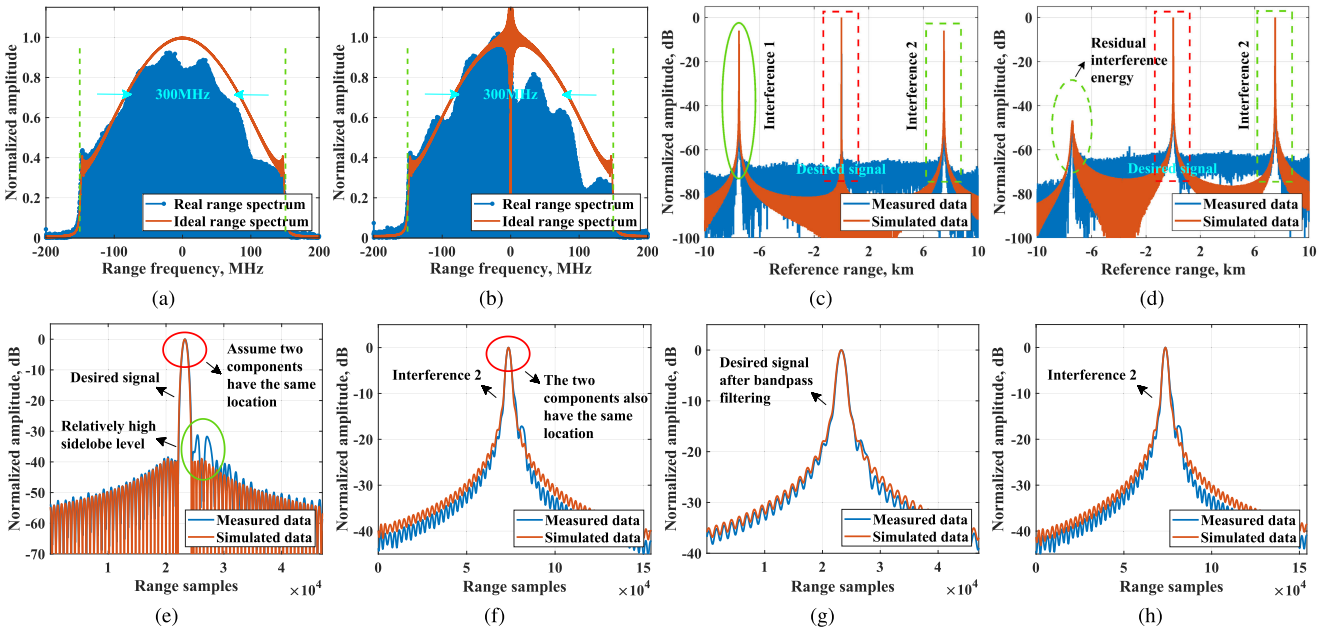


Fig. 15. Processing results of the real data recorded by the NUAA-MIMO-SAR system, as well as a comparison between the simulations and the measurements. With L=1 and M_s=2 in (10), two waveforms are transmitted by the designed system. (a) Range spectrum of $\hat{s}_1(t)$, (b) range spectrum of $\hat{s}_2(t)$, (c) match filtering results of the aliased signal returns in the current STSO scheme, and (d) match filtering results after bandpass filtering in the proposed scheme. (e)–(h) Enlarging and upsampling results of the area marked by the red rectangle in (c), the area marked by the green rectangle in (c), the area marked by the red rectangle in (d), and the area marked by the green rectangle in (d), respectively.

TABLE VI

DEVICE METRICS OF THE DESIGNED MIMO SYSTEM

Device	Metric	Value
Generation and collection of radar signals	ADC or DAC	3.2GHz
	Reference clock	10MHz
	Signal bandwidth	≤400MHz
	Number of Tx channels	2
	Number of Rx channels	2
Up-converter	Frequency of input signals	1.3∼2.0GHz
	Frequency of output signals	8.0∼12GHz
	Conversion gain	-75∼35dB
	Number of input channels	4
	Number of output channels	4
Down-converter	Frequency of input signals	8.0∼12GHz
	Frequency of output signals	1.3∼2.0GHz
	Conversion gain	65∼85dB
	Number of input channels	4
	Number of output channels	4
Tx antenna	Response range	8.2∼12.4GHz
	Antenna gain	19.4∼22.0dB
Rx antenna	Response range	8.0∼12.5GHz
	Antenna gain	19.5∼20.7dB

TABLE VII

GROUND EXPERIMENT PARAMETERS

Parameter	Value
Bandwidth	300 MHz
Sampling frequency	400 MHz
Pulse duration	100 us
Center frequency	10.0 GHz
PRF	3052 Hz
Azimuth beamwidth	20°
Elevation beamwidth	8°
Number of Tx antennas	2
Number of Rx antennas	2

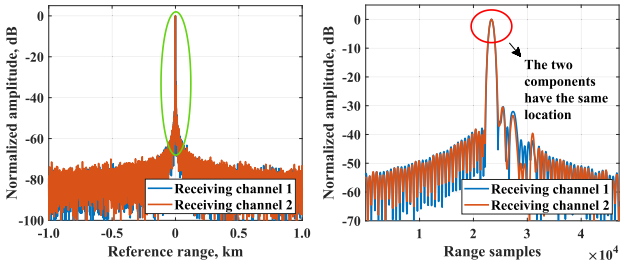


Fig. 16. Location information of the signals from different receiving channels. (a) Match filtering results of the dual-channel data. (b) Enlarging and upsampling results of the area marked by an ellipse in Fig. 16(a).

bandpass filtering in the proposed scheme, which can also be verified in the result shown in Fig. 15(d). To provide insight into the synchronization performance of the transmitters, the areas marked by rectangles in Fig. 15(c) and (d) are enlarged and upsampled, with the corresponding results being given in Fig. 15(e)–(h), respectively. In the designed system, if the STSO waveforms are simultaneously transmitted by the Txs, the location relationship between the desired signal and the interferences in the measured data must be consistent with that in the simulation, as can also be seen in Fig. 15(e)–(h). It is worth noting that the enlarging and the upsampling for interference 1 can be avoided as the time delay gap between interferences 1 and 2 is constant, being equal to T_p.

Furthermore, to illustrate the receiving synchronization of the designed system, one Tx transmits the NLFM signal with

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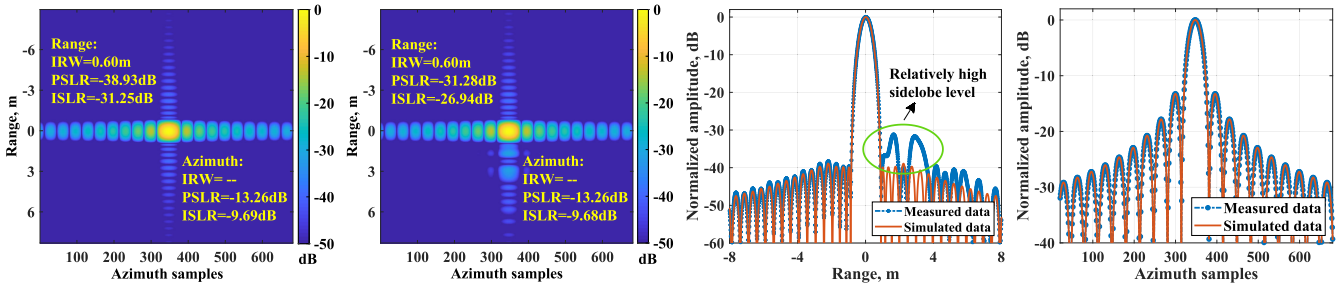


Fig. 17. 2-D coherent performance of the designed system. (a) Simulated data, (b) measured data, (c) range slices of simulated and measured data, and (d) azimuth slices of the simulated and measured data. It is worth noting that the range compression is achieved by match filtering, while the azimuth compression is performed by FFT across the azimuth direction.

two receivers recording the returned signals. In the same time segment, the raw data in the two channels are recorded and match filtered, with the processed results being shown in Fig. 16. By combining Fig. 16(a) and (b), one may conclude that the signals collected by the different channels have the same position, illustrating the synchronization of the two receivers in the designed system.

Finally, the range and azimuth coherent performance of the designed system is evaluated, with results shown in Fig. 17. Fig. 17(a) and (b) shows the 2-D coherent result from the simulation and measurement, respectively. Fig. 17(c) and (d) shows the corresponding range and azimuth slices, respectively. It can be observed that this system has better azimuth coherent performance, which is consistent with the simulation results. However, in the measured data, relatively high sidelobe may be seen to occur in the range direction [also seen in Fig. 15(e)]. Generally, the PSLR is lower than -30 dB, which will not affect the subsequent imaging performance [31]. To summarize, as indicated by the results from the measured data, the designed MIMO system has satisfactory transmitting/receiving synchronization and 2-D coherent processing capabilities.

VI. Conclusion

The separation of radiated waveforms using time and frequency synchronization constitutes a considerable challenge for the realization of the MIMO concept in SAR systems. In particular, the system resources that spaceborne/airborne MIMO-SAR can provide are usually limited for echo separation. In this article, we show that the combination of interpulse phase coding and beamforming on receive in elevation has the potential to substantially improve the capabilities and performance of MIMO-SAR imaging. In addition, compared to the existing methods, the proposed method can halve the number of interference components by fast time spectrum preprocessing and segment synthesis, which consequently saves the elevation DOFs in the beamforming operation, which is of notable importance to resource-limited MIMO-SAR remote sensing systems. Moreover, with the proposed waveform transmitting framework, most monostatic radar waveforms (e.g., NLFM signal) can be exploited to optimize imaging performance. Finally, detailed simulations on point-like targets and distributed scenes are carried out, showing that the proposed method is able to offer better echo separation performance with limited system resources, indicating the practical potential of the system.

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